Semi-Autonomous Indoor Fire-Extinguishing Drone

Manual flight + autonomous fire detection + autonomous nozzle targeting + automatic water discharge

Designed for a controlled indoor college demonstration using a small target fire such as a candle or a very small paper fire in a fire-safe metal tray.

Project objective

The existing Vikas drone is flown and positioned entirely by a human operator. A lightweight ESP32-CAM on the drone streams video to a laptop. The laptop runs the trained fire-detection model. Once a valid fire is confirmed, software calculates the fire target point, commands a two-axis nozzle to aim using a calibrated camera-to-nozzle mapping, and automatically activates a small water pump when all discharge conditions are satisfied.

What is manual and what is autonomous

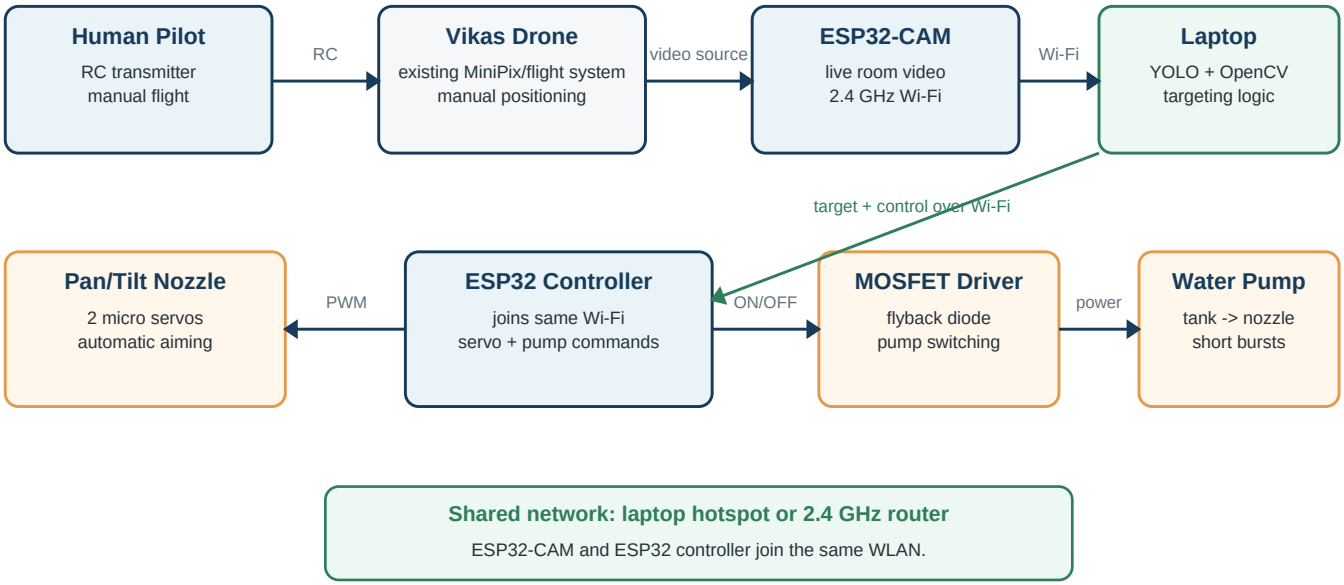
Function	Mode
Takeoff, movement, approach, hover and landing	MANUAL - human pilot
Fire detection in the camera image	AUTONOMOUS - AI on laptop
Drone positioning relative to the target	MANUAL - human pilot
Fine nozzle pan/tilt aiming	AUTONOMOUS - calibrated servo control
Water pump activation and stopping	AUTONOMOUS after arming, confirmation and predicted alignment
Emergency override / suppression arming	MANUAL safety control

Scope decision

This version deliberately avoids autonomous indoor navigation. The AI does not fly the drone and does not command the drone motors. The Vikas flight controller remains responsible only for normal manually piloted flight. This keeps the project focused on computer vision, automatic targeting, embedded control and water actuation.

Existing equipment assumed: Vikas drone, RC transmitter/receiver, flight battery/charger, and a laptop. These are excluded from the budget.

1. How the system works



The flight controller is not commanded by the AI. The ESP32-CAM and the ESP32 nozzle/pump controller should join the same 2.4 GHz Wi-Fi network, normally a laptop hotspot or a nearby router.

Simple operating sequence

- **1. Manual flight:** The pilot takes off and searches/positions the Vikas drone using the normal RC transmitter.

- **2. Video streaming:** The ESP32-CAM sends live room video over the shared Wi-Fi network to the laptop.

- **3. Autonomous fire detection:** YOLO/OpenCV detects the candle or small paper flame and produces a bounding box plus confidence score.

- **4. Manual positioning:** The pilot brings the drone to a repeatable, safe stand-off distance and keeps it roughly facing the target. The AI never moves the aircraft.

- **5. Autonomous nozzle targeting:** Software selects a target point (normally near the lower-center/base region of the detected flame) and uses the calibrated camera/nozzle geometry and distance to calculate pan/tilt servo angles.

- **6. Automatic discharge:** After the fire persists for several frames, the servos have settled, predicted aim error is within tolerance, and the operator has armed the system, the controller starts the pump automatically for a short burst.

- **7. Re-check:** The pump stops, detection is checked again, and another burst is allowed only if the fire is still valid and all safety conditions remain true.

Fail-safe rule: loss of the Wi-Fi control message, loss of valid fire detection, operator DISARM, or controller watchdog timeout must force the pump OFF.

2. Recommended technical design

The project can be built as three independent modules. Build and validate each on the bench before installing it on the drone.

A. Vision and communication module

- **ESP32-CAM:** lightweight camera on the drone; streams live video to the laptop over 2.4 GHz Wi-Fi.

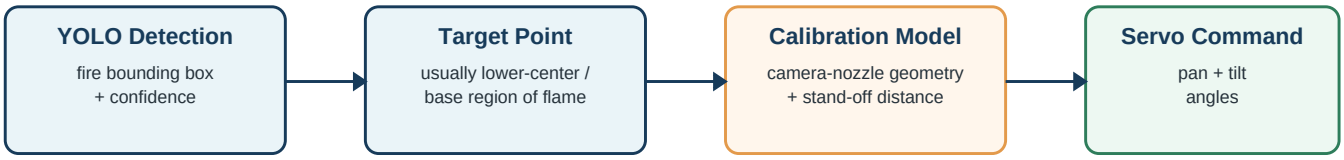
- **Laptop:** runs Python, OpenCV and the trained fire-detection model. A model fine-tuned for small indoor flames is preferred.

- **Shared WLAN:** ESP32-CAM and ESP32 controller join the same laptop hotspot/router. Avoid an architecture where each board creates a separate access point.

- **Detection confirmation:** require persistent detection across multiple frames to reject lights, reflections and orange/red objects.

- **RF coexistence test:** because the RC link may also use 2.4 GHz, verify control range, failsafe behavior and video/control stability with both ESP32 devices transmitting before indoor flight.

B. Nozzle targeting module



Important: moving the nozzle does NOT move the fire in the camera image.

Therefore alignment is based on a calibrated mapping from target pixel + distance to servo angles.

A ToF sensor improves repeatability; without it, operate near one calibrated stand-off distance.

- **ESP32 DevKit:** receives target/servo and pump commands over Wi-Fi.

- **Two micro servos:** one pan and one tilt axis. Metal-geared micro servos are preferable for repeatable positioning.

- **Distance dependence:** nozzle calibration changes with target distance. Either operate near one calibrated stand-off distance or use a small distance sensor to select the correct calibration.

- **Target point:** do not simply aim at the visual center of the flame. Use a calibrated lower-center/base-region target appropriate to the candle or paper source.

3. Water-discharge, power and safety logic

C. Water-discharge module

- **Mini DC pump:** moves water from a small reservoir through tubing to the nozzle.

- **Logic-level MOSFET + flyback diode:** the ESP32 controls the MOSFET; the pump must never be powered from an ESP32 GPIO pin.

- **Servo power:** do not power the pan/tilt servos from the ESP32 board regulator. Use an appropriately rated BEC/buck supply for the servos and electronics.

- **Grounding:** controller and servo/pump driver signals need a correct common reference ground where required. Keep pump current paths and wiring away from the camera/controller to reduce electrical noise.

- **Bench current test:** measure pump stall/start current and simultaneous servo movement before connecting the payload to the drone battery.

Suggested automatic discharge conditions

Condition	Purpose
Suppression system ARMED by operator	Prevents unexpected automatic water release
Fire persists for consecutive frames	Rejects momentary false detections
Confidence above calibrated threshold	Rejects weak detections
Target distance within calibrated range	Keeps nozzle geometry valid
Servo command / predicted impact error within tolerance	Ensures the nozzle should intersect the target region
Servos settled for a short period (for example ~0.5 s)	Avoids spraying while the nozzle is moving
Control link/watchdog healthy	Pump turns off if communication is lost
Short burst then re-check	Reduces water use and allows verification

Distance sensor recommendation: A small ToF sensor is strongly recommended for repeatable aiming, but it measures the physical surface near/behind the flame rather than the flame itself. If no sensor is used, the pilot should position the drone near one experimentally calibrated stand-off distance.

Operator arming / emergency stop

Prefer a spare RC receiver channel as the ARM/DISARM input to the suppression controller if a suitable channel is available. DISARM must override every software command and force the pump OFF immediately. If an RC channel cannot be used, provide another operator-accessible emergency-stop method; do not rely only on a switch mounted on the airborne payload.

4. Indoor payload and weight plan

The project information states a rated payload capacity of about 500 g. Do not design to the absolute limit. The actual safe margin depends on the exact battery, mounting, center of gravity and the manufacturer's test configuration. For the prototype, target an added payload of approximately **300-350 g or less** and verify final all-up mass and flight behavior.

Conservative example payload

Item	Approx. mass
ESP32-CAM	10-20 g
ESP32 controller	10-20 g
2 x micro servos	20-30 g
Pan/tilt bracket + nozzle linkage	20-35 g
Mini pump	25-50 g
MOSFET/BEC/wiring	25-45 g
Light reservoir + tube + nozzle	30-50 g
Mounting hardware	25-50 g
Water (75-100 mL)	75-100 g
Estimated total added payload	~240-400 g depending on hardware; target <=300-350 g

Recommended water quantity

Start with roughly **75-100 mL** for a candle/small-paper-fire demonstration. One millilitre of water has approximately one gram of mass, so water quickly becomes a large fraction of the payload. Increase only after bench testing shows it is necessary.

Mounting guidance

- Place the reservoir close to the aircraft center of gravity and prevent it from shifting or sloshing significantly.

- Keep the camera and nozzle physically close to reduce short-range parallax, while still calibrating the actual offset.

- Keep water lines away from flight electronics and protect electrical joints from leaks.

- Test with solid ballast before flying with water. Check stability, hover throttle, battery time and motor/ESC temperature after each payload increase.

- Do not assume the nominal 500 g payload rating guarantees safe indoor flight at 500 g in the exact battery/configuration being used.

Before live-fire testing, verify the complete flight-ready drone with the actual battery and mounted payload. A stable center of gravity and adequate thrust margin matter more than the nominal payload number.

5. Conservative budget (drone and laptop already available)

The following budget intentionally uses rounded, conservative amounts rather than the lowest online price. It allows for better-quality parts, shipping, connectors, replacements and mounting changes.

Item	Qty	Conservative budget	Notes
ESP32-CAM camera module	1	Rs 800	Video streaming to laptop
ESP32 DevKit controller	1	Rs 500	Servo + pump controller
MG90S-class micro servos	2	Rs 400	Pan and tilt
Pan/tilt bracket + nozzle linkage	1	Rs 300	Bought or fabricated
Mini DC water pump	1	Rs 300	Small low-voltage pump
Logic-level MOSFET + diode + terminals	1 set	Rs 250	Pump switching/protection
BEC/buck converter(s)	1-2	Rs 500	Adequate current for electronics/servos
Reservoir, tubing and nozzle	1 set	Rs 600	Lightweight tank and water path
Wires, connectors, perfboard, RC/safety interface	1 set	Rs 600	Integration + ARM/DISARM path
Mounting / 3D print / fasteners	1 set	Rs 600	Camera, controller, tank and nozzle
Spare/repair allowance	-	Rs 700	Extra servo/pump/cables etc.
Shipping / miscellaneous	-	Rs 500	Conservative allowance
Core project subtotal	-	Rs 6,050	Excludes drone and laptop
Recommended ToF distance sensor	1	Rs 500-1,000	Improves repeatable nozzle calibration
Optional dedicated 2.4 GHz router/travel AP	1	Rs 0-2,000	Only if laptop hotspot/existing router is unsuitable
Maximum recommended project allocation	-	Rs 8,000-10,000	Includes contingency and optional items

Expected spend for the first working prototype: about Rs 5,500-7,000. Keep up to Rs 8,000-10,000 available only if replacements, better mounts, a distance sensor or a dedicated network device become necessary.

Budget notes

- No GPS, optical flow, autonomous flight controller upgrade or MAVLink telemetry is required for this revised project scope.

- The ToF sensor is not strictly mandatory, but it is now recommended because nozzle calibration is distance-dependent.

- Do not choose the cheapest pump/servo only by price; actual mass, current draw, repeatability and water-jet range matter.

- A laptop hotspot may cost nothing. Add a small router only if connection stability during testing is poor.

6. Suggested build, calibration and test sequence

- **Vision bench test:** stream ESP32-CAM video to the laptop and verify candle/small-paper flame detection at intended room distances and lighting.

- **Network test:** connect both ESP32 boards to the same WLAN and verify simultaneous video streaming and low-latency command delivery.

- **RC coexistence test:** with props removed or aircraft restrained, verify normal RC behavior and failsafe while Wi-Fi video/control traffic is active.

- **Nozzle calibration:** with the drone powered off, place the camera/nozzle assembly at known distances and record the servo pan/tilt angles that make the water impact the desired base-region target. Store a lookup table or fitted mapping.

- **Dry targeting test:** detect the flame and command the servos using the calibration model without running the pump.

- **Static water test:** verify water impact at several distances and adjust the calibration. Confirm that pump start/stop does not reset the ESP32 or disturb servos.

- **Fail-safe test:** DISARM, loss of target, Wi-Fi loss and watchdog timeout must each turn the pump OFF.

- **Payload flight test:** fly with equivalent solid ballast before adding water; then add the final hardware and only a small water quantity.

- **Controlled integrated demo:** pilot manually to the calibrated stand-off area, allow autonomous detection/aiming, ARM the suppression system, and allow automatic short-burst discharge after conditions are met.

Indoor demonstration safety

- Use a large, clear indoor test area with a trained pilot and keep spectators well outside the flight path. Exposed propellers on a research quadcopter are a major indoor hazard.

- Prefer a stable candle/tea-light in a non-flammable holder for early integrated tests. If paper is used, keep the quantity very small and inside a metal tray or other fire-safe container.

- Rotor downwash can extinguish a candle or spread burning paper before the water reaches it. Establish the stand-off distance experimentally and never hover directly above the flame.

- Keep a separate conventional extinguisher or water source available, and have a second person supervise the live flame.

- Do not use water on energized electrical equipment, flammable-liquid/grease fires, batteries, or any target where water is inappropriate.

- Initial aiming, pump and calibration tests should be done with the aircraft powered off or securely restrained before flight tests.

Final project definition

A manually flown indoor Vikas drone that autonomously detects a small fire, automatically aims a two-axis water nozzle using calibrated target geometry, and automatically activates the water pump in short bursts after target confirmation, predicted alignment and operator arming.

This scope is practical for a college project because the pilot remains responsible for aircraft navigation and positioning, while the system demonstrates the key intelligent functions: vision-based fire detection, calibrated automatic nozzle targeting and autonomous water actuation. The final report should document detection accuracy, targeting error versus distance, discharge latency, false-positive behavior and payload/flight-time impact.

Revision summary: This revised version corrects the Wi-Fi topology, makes nozzle calibration explicitly distance-dependent, clarifies that nozzle movement does not move the fire within the fixed camera image, strengthens servo/pump power guidance, adds 2.4 GHz coexistence testing, adds link-loss fail-safe behavior, and refines the fire target point to the lower/base region.